KHETAM MOHAMED

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PROFESSIONAL SUMMARY

Computer Science graduate (Artificial Intelligence track, Nile University, GPA 3.8/4.0) who builds applied AI systems end to end. Built an agentic AI system orchestrating Gemini 2.5 Flash and a fine-tuned Qwen3-8B with a RAG pipeline, Egyptian Arabic TTS/STT modules and a FastAPI backend. Hands-on experience in NLP, OCR, computer vision (PyTorch, TensorFlow, OpenCV) and MLOps (DVC, MLflow, Docker). Four competition placements, including two 1st places.

EDUCATION

B.Sc. in Computer Science, Artificial Intelligence Track | Nile University 2022 – 2026
GPA 3.8/4.0 | Dean's Honor List, 2026 | Coursework: Machine Learning, NLP, Computer Vision, Data Structures and Algorithms

TECHNICAL SKILLS

Generative AI & LLMs: LLM fine-tuning, RAG, agentic AI / LLM orchestration, prompt engineering
Models & Architectures: Qwen3-8B (fine-tuned), Gemini 2.5 Flash (API), ResNet50 + U-Net, GANs, Random Forest, XGBoost
ML & Deep Learning: Supervised/unsupervised learning, CNNs, RNNs, LSTM, model evaluation, data augmentation
NLP, Speech & Vision: NLP, information extraction, sentiment analysis, OCR, TTS/STT, image segmentation, super-resolution
Programming & Libraries: Python, C++, Java, SQL; PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, NumPy, Pandas
MLOps & Backend: FastAPI, REST APIs, Docker, DVC, MLflow, DagsHub, Prometheus, Git, ETL pipelines, MySQL, SQLite, MongoDB

PROJECTS

BASEET: Agentic AI and Sensor-Integrated Solution for Adaptive Special Education Graduation Project | 2026
Adaptive learning system for special-education students that personalizes content and responds to sensor-detected stress.
- Built an agentic AI system orchestrating 2 models, Gemini 2.5 Flash and a fine-tuned Qwen3-8B, with a RAG pipeline to deliver personalized learning content for special-education students.
- Developed a Random Forest classifier on real-time sensor data to detect behavioral and stress indicators and trigger adaptive intervention.
- Developed an Egyptian Arabic text-to-speech (TTS) module and contributed to the speech-to-text (STT) module to make content accessible to Arabic-speaking students.
- Delivered a FastAPI backend (REST API) coordinating 3 components: model inference, sensor input, and adaptive content delivery.

Arabic Prescription Understanding and Digitization System (OCR + NLP) 2026
Digitizes handwritten Arabic and English medical prescriptions into structured text.
- Developed the OCR and NLP pipeline, including text preprocessing and information extraction.

Telco Customer Churn Prediction: End-to-End MLOps Pipeline
End-to-end MLOps pipeline for predicting telecom customer churn.
- Contributed the statistics and analysis that benchmarked 5 models in a pipeline using DVC, MLflow, DagsHub, Docker and Prometheus (deployed on Azure Container Apps); tested the FastAPI service under load.

GAN-Based Super-Resolution for CCTV Enhancement 2024 – 2025
Enhances low-resolution CCTV footage for face and object recognition.
- Built the GAN super-resolution pipeline (PyTorch, OpenCV) with denoising, improving recognition by approximately 30%.

Multi-Task Breast Cancer Detection using ResNet50-U-Net
Analyzes breast ultrasound images to segment lesions and classify the diagnosis.
- Built a multi-task deep learning framework (pre-trained ResNet50 encoder + U-Net) with CLAHE preprocessing and test-time augmentation.

AWARDS AND ACHIEVEMENTS

- **1st Place, NLP Track** – 21st Undergraduate Research Forum (UGRF), 2026: Arabic Prescription Digitization System
- **2nd Place, IoT Track** – 21st Undergraduate Research Forum (UGRF), 2026: Autism Stress Tracking Assistant
- **5th Place, National Undergraduate Research Forum** – 2025: GAN-Based Super-Resolution for CCTV Enhancement
- **1st Place, AI Agent Development Competition** – collaborated on building an AI assistant agent

EXPERIENCE AND LEADERSHIP

Junior Teaching Assistant, Python Programming | Nile University 2023 – 2024
- Supported Python labs and debugging sessions for 20–30 students; graded assignments and quizzes with feedback.

Head, Google Developer Groups (GDG) Club | Nile University Oct 2024 – Oct 2025
- Led the club's student team and promoted its technical courses to students.

Languages: Arabic (Native) | English | French